

- 30.** Prove that right- and left-cancellation hold for a group G . That is, prove that when $a, b, c \in G$, then $ba = ca$ implies $b = c$, and $ab = ac$ implies $b = c$.

Proof. Suppose $a, b, c \in G$, and $ba = ca$. Multiply both sides by a^{-1} on the right to get

$$(ba)a^{-1} = (ca)a^{-1}.$$

Using associativity, this becomes

$$b(aa^{-1}) = c(aa^{-1})$$

$$be = ce$$

$$b = c.$$

Thus we have shown $ba = ca$ implies $b = c$.

Now suppose $ab = ac$. Multiply both sides by a^{-1} on the left to get

$$a^{-1}(ab) = a^{-1}(ac).$$

Next use associativity to get

$$(a^{-1}a)b = (a^{-1}a)c$$

$$eb = ec$$

$$b = c.$$

Thus we have shown $ab = ac$ implies $b = c$. ■

- 32.** Prove that if G is a group of even order, then there is an element $a \in G$, with $a \neq e$, and $a^2 = e$.

Proof. (Contrapositive) Suppose that there is no element $a \in G$ for which $a \neq e$ and $a^2 = e$. Thus, for each non-identity element $a \in G$, we have $a^2 \neq e$, which is to say $aa \neq e$. This means that $a^{-1} \neq a$. Consequently any $a \in G$ (other than e) has an inverse that is unequal to a .

Thus the non-identity elements of G can be grouped in pairs a and a^{-1} . In fact, imagine listing the non-identity elements of G in a table as follows, so each column contains a particular element $a_i \in G$ and its inverse a_i^{-1} , and every element of G (other than e) appears exactly once in the table.

a_1	a_2	a_3	a_4	$\dots$	a_n
a_1^{-1}	a_2^{-1}	a_3^{-1}	a_4^{-1}	$\dots$	a_n^{-1}

It follows that G has an even number $2n$ of non-identity elements. But, in addition, G has the identity element e . Therefore G has a total of $2n + 1$ elements. Consequently G has odd order. ■

- 33.** Let G be a group and suppose $(ab)^2 = a^2b^2$ for each $a, b \in G$. Prove that G is abelian.

Proof. Suppose G is a group and $(ab)^2 = a^2b^2$ for each $a, b \in G$. Take any two elements $a, b \in G$. Then we have $(ab)(ab) = a^2b^2$, that is $abab = aabb$, which is $a(bab) = a(abb)$. Cancellation (Exercise 28 above) gives

$$bab = abb,$$

or $(ba)b = (ab)b$. Cancellation again gives $ba = ab$. We have now shown that $ba = ab$ for any $a, b \in G$. This means G is abelian. ■

34. Find all subgroups of $\mathbb{Z}_3 \times \mathbb{Z}_3$. Deduce that $\mathbb{Z}_3 \times \mathbb{Z}_3$ is not the same as $\mathbb{Z}_9$.

The subgroups are as follows:

$$H_1 = \{(0, 0)\}$$

$$H_2 = \{(0, 0), (1, 0), (2, 0)\}$$

$$H_3 = \{(0, 0), (0, 1), (0, 2)\}$$

$$H_4 = \{(0, 0), (1, 1), (2, 2)\}$$

$$H_5 = \{(0, 0), (1, 2), (2, 1)\}$$

$$H_6 = \{(0, 0), (1, 0), (2, 0), (0, 1), (1, 1), (2, 1), (0, 2), (1, 2), (2, 2)\}$$

By contrast, $\mathbb{Z}_9$ has only three subgroups:

$$H_1 = \{0\}$$

$$H_2 = \{0, 3, 6\}$$

$$H_3 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}.$$

Therefore $\mathbb{Z}_3 \times \mathbb{Z}_3$ and $\mathbb{Z}_9$ have different structures and are not the same.

41. Prove that $G = \{a + b\sqrt{2} : a, b \in \mathbb{Q} \text{ and } a \text{ and } b \text{ are not both } 0\}$ is a subgroup of $\mathbb{R}^*$ under the group operation of multiplication.

Certainly we have $G \subseteq \mathbb{R}^*$. We will apply Proposition 3.9 to show that G is a subgroup of $\mathbb{R}^*$.

1. The identity 1 of $\mathbb{R}^*$ has form $1 = 1 + 0\sqrt{2} \in G$, so $1 \in G$.
2. Consider two elements $a + b\sqrt{2}$ and $a' + b'\sqrt{2}$ in G , so $a, b \in \mathbb{Q}$ and are not both zero, and likewise for a' and b' . Their product is $(a + b\sqrt{2})(a' + b'\sqrt{2}) = (aa' + 2bb') + (ab' + ba')\sqrt{2}$, and this has the required form $x + y\sqrt{2}$, where x and y are rational. (As $x = aa' + 2bb'$ and $y = ab' + ba'$ are products and sums of rational numbers, they are themselves rational.) Moreover, $x = (aa' + 2bb')$ and $y = (ab' + ba')$ are not both zero, for otherwise the product $(a + b\sqrt{2})(a' + b'\sqrt{2}) = x + y\sqrt{2}$ of two nonzero elements of $\mathbb{R}^*$ would be zero, which is impossible. Therefore the product $(a + b\sqrt{2})(a' + b'\sqrt{2})$ is in G .
3. Consider an arbitrary element $a + b\sqrt{2}$ in G . Its inverse in $\mathbb{R}^*$ is

$$(a + b\sqrt{2})^{-1} = \frac{1}{a + b\sqrt{2}} = \frac{1}{a + b\sqrt{2}} \frac{a - b\sqrt{2}}{a - b\sqrt{2}} = \frac{a}{a - 2b} + \frac{-b}{a - 2b}\sqrt{2} \in G$$

Observations 1–3 above combined with Proposition 3.9 prove that G is a subgroup of $\mathbb{R}^*$. ■

42. Show that $H = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a + d = 0 \right\}$ is a subgroup of the group $G = \mathbb{M}_2(\mathbb{R})$ of 2×2 matrices under matrix addition.

Certainly we have $H \subseteq G$. We will apply Proposition 3.9 to show that H is a subgroup of G .

1. The additive identity matrix $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ is in H because $a + d = 0$ for this matrix.
2. Consider two matrices $H_1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $H_2 = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ in H , so $a + d = 0$ and $a' + d' = 0$. Observe that $H_1 + H_2 = \begin{pmatrix} a + a' & b + b' \\ c + c' & d + d' \end{pmatrix}$ satisfies $(a + a') + (d + d') = (a + d) + (a' + d') = 0$, so $H_1 + H_2 \in H$.
3. Consider an arbitrary element $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in H , so $a + d = 0$. The inverse of this matrix is $\begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix}$, and as $(-a) + (-d) = -(a + d) = 0$, this inverse is in H .

Observations 1–3 above combined with Proposition 3.9 prove that H is a subgroup of G . ■

45. Prove that the intersection of two subgroups of a group is also a subgroup.

Proof. Suppose H and K are two subgroups of a group G . In what follows, we use Proposition 3.9 to show that $H \cap K$ is a subgroup of G .

1. Since H is a subgroup of G , we must have $e \in H$, by definition of a subgroup. For the same reason $e \in K$. Therefore, $e \in H \cap K$, by definition of intersection.
2. Suppose $h_1, h_2 \in H \cap K$. Then $h_1, h_2 \in H$ and $h_1, h_2 \in K$ by definition of intersection. But then we have $h_1 h_2 \in H$ and $h_1 h_2 \in K$ since H and K are subgroups (and are closed under the group operation). Thus $h_1 h_2 \in H \cap K$ by definition of intersection.

We've now shown that whenever h_1 and h_2 are in $H \cap K$, the product $h_1 h_2$ is also in $H \cap K$.

3. Suppose $h \in H \cap K$. Then $h \in H$ and $h \in K$, by definition of intersection. Therefore $h^{-1} \in H$ and $h^{-1} \in K$ because H and K are subgroups. Consequently $h^{-1} \in H \cap K$.

We've now shown that whenever h is in $H \cap K$, the inverse h^{-1} is also in $H \cap K$.

Observations 1–3 above combined with Proposition 3.9 prove that $H \cap K$ is a subgroup of G . ■

51. If $xy = x^{-1}y^{-1}$ for every $x, y \in G$, then G is abelian.

Proof. Suppose $xy = x^{-1}y^{-1}$ for every $x, y \in G$. Then given any $x \in G$, we can set $y = e$, and get $xe = x^{-1}e^{-1}$, which yields $x = x^{-1}$. Thus $xx = e$ for each $x \in G$.

Now take any $a, b \in G$. By the above, we must have $(ab)(ab) = e$, or $abab = e$. Left-multiply both sides of this by a to get $aabab = a$. Now right-multiply both sides of *this* by b to get $aababb = ab$, which is $(aa)ba(bb) = ab$. Using the fact that $aa = e$ and $bb = e$, this becomes $ba = ab$. It follows that G is abelian. ■